

Vuka: DSAC Public Entities Reporting — PRD

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1. Problem and context

The Department of Sport, Arts and Culture funds 26 public entities and six non-profit organisations. Each is obliged to report back to DSAC on two things: performance against agreed targets, and accountability for the funds received.

There is no digital system for this. Reporting happens by email, in documents written to whatever format each entity chooses.

Three consequences follow, all stated in the challenge brief.

Entities miss deadlines. Manual reporting over email has no enforcement, no visibility of what is outstanding, and no early signal that a submission is at risk.

DSAC cannot analyse what arrives. The reports come in inconsistent formats and the review process is entirely manual. Comparing one entity against another, or this year against last, means a person reading documents side by side. Management information is not available at the click of a button; it is not available at all without weeks of work.

The public cannot see what the money produced. The brief makes this point explicitly: the absence of a system to track allocation and usage of funds, and reporting on service delivery obligations to citizens, affects public sentiment around the services these institutions render.

That third consequence is the one most likely to be treated as background colour by other teams. We treat it as the centre of the problem. The reporting failure is not merely an administrative inconvenience inside a department. It is a break in the line between public money going out and a citizen being able to see what it bought.

2. Product concept: the accountability chain

Working name: Vuka (to wake up, to rise). Placeholder, open to change.

One line. Vuka traces every rand DSAC allocates through to the target it funded, the evidence that the target was met, and a public record of what it produced.

The brief asks for a reporting system. A reporting system moves documents from entities to a department and shows a dashboard. That is the floor, and we will build it. But a reporting system solves only the first of the three problems above.

The product concept is a chain with four links, each of which must connect to the next:

Allocation. What each entity received this financial year, and against which programme.

Targets. What that entity committed to deliver for the money, drawn from its Annual Performance Plan.

Evidence. What it actually reported, quarter by quarter, with the source document behind every figure.

Outcome. What the public can see: allocation, commitment, delivery, and how recently the entity reported.

When the chain is intact, several things become possible that are impossible today. Cost per outcome can be compared across entities. Under-delivery becomes visible mid-year rather than at year end. And the department can answer a journalist or a parliamentary question in minutes.

What this is not

Not a document management system. Documents are an input, not the product. If the system's value were storing PDFs, SharePoint already exists.

Not an audit or assurance tool. We surface signals and route them to a human reviewer. We do not certify performance information; that remains the Auditor-General's role.

Not an autonomous decision-maker. Every AI-derived figure and every risk score is a proposal shown with its reasoning, confirmed by a named human before it counts.

3. Users

Primary: the entity reporter

Finance or strategy staff at one of the 26 public entities, or at one of the six NPOs. The two groups differ sharply and the product must serve both.

A large entity has a finance team, a planning unit, reliable connectivity and existing internal reporting rhythms. Its problem is duplication: it produces substantially the same information for DSAC, National Treasury, the Auditor-General and its own board, in four different shapes.

An NPO may be two or three people. Its problem is capacity: reporting competes directly with delivering the service. Connectivity may be a phone on mobile data. For this user, any system that demands an hour at a desktop portal will be used late or not at all.

What both need: to know exactly what is outstanding, to submit without rekeying work they have already done, and to know where they stand before the department tells them.

Primary: the DSAC oversight official

Sits in the department, responsible for monitoring performance across all 32 funded bodies. Today they read documents and build spreadsheets by hand.

What they need: to know which entity needs attention this week and why, to see any figure's source without asking for it, to compare entities on a like-for-like basis, and to produce management information on demand rather than on request.

Secondary: the DSAC executive

The DG, deputy DGs, and anyone who must answer to Parliament, the media or the Minister. Consumes summaries, not screens.

What they need: a short, current, defensible picture of portfolio performance, and the ability to trace any statement in it back to a source.

Secondary: the citizen

A member of the public, a journalist, a researcher or a civil-society organisation. No account, no training, often on a low-end phone and limited data.

What they need: to see what a given entity received, what it promised, what it delivered, and when it last reported, in plain language.

4. Scope: the brief's mandatory requirements

These five come directly from the challenge statement. They are worth 20 percent of the score on their own (Relevance) and underpin Technical Feasibility (20 percent). They are the floor. Everything in section 5 sits on top of them and scores nothing if the floor is missing.

a. Analytics module. A dashboard per public entity showing trends and patterns, including audit findings and year-on-year comparisons. Annual targets shown by state: in progress, not started, deadline missed. Statistics on staff demographics and jobs created per entity.

b. Early warning. Risk-based alerts drawing on reporting and historical information to identify potential delay or non-performance before deadlines are missed. Notifications on regulatory due dates and closing dates, with countdowns at 30 days, 15 days, and hourly as the deadline approaches.

c. Document repository. Storage for strategic plans, Annual Performance Plans, operational plans, annual reports, quarterly reports and financials, uploaded by entities and reviewable by DSAC.

d. Workspaces for public entities. Per-entity workspace with: integration with Microsoft technologies; version control triggered on save or upload; comments visible on screen in real time; task assignment internally and externally, up and down the operations process; approval confirmation on receipt of upload; and full functionality on mobile devices.

e. Security and privacy. Demonstrated alignment with South African cybersecurity and privacy principles. Covered in section 9.

Known scope risks

Two items in (d) are disproportionately expensive relative to their weight in the marking.

Microsoft integration. Full Microsoft 365 integration (SharePoint or Graph document sync, Teams presence) is not achievable in the build window. Position: demonstrate Office document handling and Entra ID single sign-on as the integration path, and present the deeper integration as roadmap with a credible technical approach. The brief's judging criteria explicitly allow partial implementation where the approach is credible.

Real-time comments. Live collaborative commenting requires a websocket layer that is custom work in this stack. Position: build threaded comments anchored to specific targets and documents, with near-real-time refresh. Comment-per-target is arguably more useful to a reviewer than a live cursor anyway, since it routes feedback to the thing being disputed.

5. Differentiating features

Five features that no other team is likely to build. Each maps to a scoring criterion beyond mere relevance.

5.1 Rand per outcome

Every dashboard at this hackathon will show targets in red, amber and green. Ours leads with a different number: cost per outcome. Cost per job created, cost per target achieved, cost per beneficiary reached, comparable across all 32 funded bodies and against the same entity's prior years.

This answers the question the department actually has. Not *did they report*, but *was the money worth it*. It also makes peer benchmarking fall out for free: an entity delivering at three times the portfolio average cost per job is visible immediately.

Caveat to state openly in the pitch: outcomes are not always commensurable across a ballet company and a sports federation. The comparison is a prompt for a conversation, not a verdict, and the UI should say so.

5.2 Report once, satisfy everyone

Entities report substantially the same underlying facts to DSAC, National Treasury, the Auditor-General and Parliament, in four different formats. That duplication is a direct cause of the missed deadlines the brief complains about.

Vuka models the facts once — target, actual, variance, spend, evidence — and generates the different output shapes from that single source. Submit to DSAC, export the Treasury view.

This is the requirement behind the requirement. It is also the feature most likely to make a real entity CFO in the room lean forward.

5.3 Document extraction: upload the report, not the form

Entities already write their APPs and quarterly reports. Asking them to rekey that into web forms is how reporting systems die.

Instead: the entity uploads the document it already wrote. The system extracts targets, actuals, variance explanations and spend figures into structured data, and presents them back for confirmation. The reporter corrects what the extraction got wrong and confirms. **Nothing enters the record until a named human confirms it.**

Every extracted figure retains a pointer to its source document and location, so a reviewer can open the page the number came from.

This converts requirement (c), the document repository, from a filing cabinet into the system's actual input mechanism.

5.4 Behavioural risk scoring

The brief asks for countdowns at 30 days, 15 days and hourly. Everyone will build that. It is a calendar, not intelligence.

On top of it, a risk score per entity derived from behaviour and history:

- Submission lateness across prior reporting periods
- Evidence gaps: targets with no supporting evidence attached, weighted by how far into the year we are
- Spend-to-delivery divergence: funds substantially drawn against targets substantially not started
- Prior audit findings, and whether they were repeat findings
- Revision churn: targets restated or reworded mid-year

The score is always shown with its contributing signals. The interface never displays a bare number. It displays: *high risk, because late in three of the last four quarters, 60 percent of the year elapsed with 20 percent of targets started, and 70 percent of allocation drawn.* A reviewer can disagree with any individual signal.

This directly addresses criterion 5's requirement for transparency and explainability, and criterion 6's requirement for human oversight.

5.5 Low-bandwidth submission

The six NPOs are small and under-resourced. A desktop portal is a barrier for exactly the bodies least able to absorb one.

A lightweight submission channel — WhatsApp or plain SMS, and a low-data mobile web view — where an NPO can send a document or answer one short prompt per target, landing in the same data model as a full desktop submission.

This is a substantive answer to criterion 4 (accessibility, low bandwidth, varying digital literacy) rather than a claim about font sizes.

5.6 Public outcome view

A public page per entity, no login: allocation received, targets committed, delivery reported, date last reported. Plain language, minimal data payload.

This is the link in the chain that closes the loop on the brief's own statement about public sentiment, and it is the clearest expression of the digital sovereignty and citizen-outcome framing in criterion 7.

6. User journeys

6.1 Entity reporter: submitting a quarterly report

Thandi handles finance at a small arts NPO.

1. **Notified, usefully.** 30 days before the quarterly deadline she gets a notification. It does not just say a report is due; it lists which targets currently have no evidence attached. At 15 days it repeats with what is still outstanding. In the final hours it escalates.
2. **Submits what she already wrote.** She uploads her quarterly report document. The system extracts targets, actuals, variances and spend, and shows the structured result back to her.
3. **Confirms or corrects.** She fixes the one line the extraction misread and confirms. Nothing counts until she does.
4. **Or submits from a phone.** Low on data, she sends the document over WhatsApp instead, or answers one short prompt per target. Same destination, same data model.
5. **Sees herself as the department sees her.** Her own dashboard shows targets by state, her submission history, and her current risk score with its reasons. No surprises when DSAC calls.
6. **Receives feedback in place.** If the reviewer disputes a figure, the comment is attached to that target, not buried in an email thread.

6.2 DSAC oversight official: the Monday review

Sipho monitors all 32 funded bodies.

1. **One screen, ranked by risk.** Entities ordered by risk score, not alphabetically. The entity whose score moved this week is at the top, with the reasons beside it.
2. **Drills into the entity.** Trend over prior years, audit findings history, targets by state, staff demographics, jobs created.
3. **Checks the source.** Every figure carries a link to the document and location it was extracted from. Nothing is taken on faith.
4. **Compares against peers.** Cost per job created here versus the portfolio average. The conversation he needs to have with the board, which today he cannot have.
5. **Reviews and responds.** Approves the submission, or returns it with a comment anchored to the specific target in dispute.

6. **Generates, rather than rebuilds.** Needs the Treasury or AG cut of the same information: generates it.

6.3 Executive: preparing for a question

The DG has a parliamentary question on an entity's performance. She opens the portfolio brief: a plain-language summary of current state across the portfolio, every claim linked to the submission and document behind it. Two minutes, defensible.

6.4 Citizen: checking an entity

A journalist opens the public page for an entity. Allocation, commitments, delivery, date last reported. Plain language, no login, loads on a low-end phone.

6.5 Demo script (90 seconds)

The demo is journey 1 flowing into 2 flowing into 4, in one unbroken thread.

Time	On screen	Said
0:00	WhatsApp on a phone, NPO submits quarterly report	"An NPO with three staff and no finance system submits from a phone."
0:15	Extraction result, targets and actuals structured, human confirms	"They upload what they already wrote. We structure it. A human confirms before anything counts."
0:35	DSAC dashboard updates live, risk score moves	"The department sees it immediately."
0:50	Risk score expanded, contributing signals visible	"And it tells you why. Late three quarters running, evidence missing, spend ahead of delivery. Not a black box."
1:10	Rand per outcome comparison across entities	"Which lets you ask the real question. Not did they report. Was the money worth it."
1:25	Public entity page	"And a citizen can see what their money bought."

Closing line, delivered separately: built on Shesha, an open-source South African framework, on ASP.NET Core and SQL Server, deployable on-premise or in any cloud. No vendor lock-in, no licence per seat, and DSAC owns it.

7. Data model

Shesha is domain-driven: entity classes come first and CRUD APIs are generated from them. Getting this model right on the first pass is most of the build.

Entity	Purpose	Key fields
PublicEntity	A funded body (26 PEs, 6 NPOs)	Name, type (PE/NPO), mandate, sector, contact person, size band
FinancialYear	Reporting year	Label (2026/27), start, end
Allocation	Money committed to an entity for a year	PublicEntity, FinancialYear, programme, amount, date approved
Target	A commitment from the APP	PublicEntity, FinancialYear, description, KPI, baseline, annual target, unit of measure, quarterly breakdown, linked Allocation
ReportingPeriod	Q1-Q4 plus annual	FinancialYear, label, period start/end, submission due date, regulatory deadline
Submission	One entity's report for one period	PublicEntity, ReportingPeriod, status (draft/submitted/under review/returned/approved), submitted date, submitted by, channel (web/mobile/WhatsApp)
TargetResult	Actual against one target for one period	Target, Submission, actual value, variance, variance explanation, spend to date, status (not started/in progress/achieved/missed)
DocumentRecord	An uploaded file	PublicEntity, document type (APP/strategic plan/quarterly/annual/financials), Submission, file, version, uploaded by, uploaded date, approval status
ExtractionResult	What was read out of a document	DocumentRecord, extracted field, extracted value, source page/location, confidence, confirmed by, confirmed date
RiskSignal	One contributing factor to a score	PublicEntity, signal type, value, weight, period observed, description
RiskScore	Computed score at a point in time	PublicEntity, ReportingPeriod, score, band, computed date, contributing RiskSignals

Entity	Purpose	Key fields
AuditFinding	Historical AG findings	PublicEntity, FinancialYear, finding type, description, repeat finding flag, resolution status
WorkforceStat	Demographics and jobs	PublicEntity, ReportingPeriod, headcount, demographic breakdown, jobs created, job type
CommentThread	Feedback anchored to something	Related object (Target/TargetResult/DocumentRecord), author, body, parent, created date, resolved
TaskItem	Assigned work	PublicEntity, assigned to, assigned by, description, due date, status, internal/external
NotificationRule	Drives countdowns	ReportingPeriod, trigger offset (30d/15d/hourly), channel, recipient role

Notes on the model

The chain is the foreign keys. Allocation → Target → TargetResult →

DocumentRecord/ExtractionResult is what makes the accountability chain more than a slogan. Rand per outcome is a query across that path.

ExtractionResult is deliberately separate from TargetResult. Machine-read values live in their own table with confidence and source location until a human confirms them into the record. This separation is what lets us claim human-in-the-loop credibly rather than rhetorically.

RiskSignal is separate from RiskScore for the same reason: the score must be decomposable on screen into the signals that produced it.

Shesha base entities. Person, Organisation and Site come with the framework. PublicEntity should extend Organisation rather than duplicate it, and reporters should be Persons linked to it.

8. Technical architecture

Platform

Shesha: an open-source low-code framework built on ASP.NET Core, ABP, React and NextJS, with SQL Server. Domain-driven, with CRUD APIs generated from entity classes and authorisation configured rather than coded. Drag-and-drop form builder with 40-plus standard components, app themer, and an administration panel covering user and security management.

Deploys like any ASP.NET Core website: on-premise or on any cloud.

What comes free

The following need configuration, not code:

- Entity CRUD and REST APIs, generated from the domain classes in section 7
- Role-based access control and the security model
- User, role and permission administration
- The document repository, including upload, versioning and approval state
- Standard list, form, detail and dashboard pages via the form builder
- Branding and theming
- Audit trails on entity changes

This is why DSAC was the right challenge to pick: requirements (c), (d) and most of (a) are largely platform features.

What we build

Custom React components for the pieces the standard component set will not cover: the risk score panel with expandable contributing signals, the rand-per-outcome comparison view, and the public entity page.

Extraction service. A .NET service that takes an uploaded document, extracts targets, actuals, variances and spend, and writes ExtractionResult rows with source location and confidence. Called from the upload flow; results surface in the confirmation screen.

Risk scoring job. A scheduled job computing RiskSignals and RiskScore per entity per period. Deterministic and rule-weighted rather than an opaque model, precisely so it can be explained on screen and defended to a judge.

Notification scheduler. Drives the 30-day, 15-day and hourly countdowns off ReportingPeriod due dates, with escalation rules.

Inbound WhatsApp/SMS channel. A webhook endpoint that accepts a document or a short structured exchange and creates a Submission through the same service layer as the web path. Same validation, same data model.

Public read-only surface. A separate, unauthenticated, cache-friendly view over a deliberately narrow projection of the data. No login, minimal payload.

Sovereignty position

Worth stating explicitly in the pitch because it maps almost verbatim onto criterion 7: open-source framework, developed in South Africa, already used in production in government. Standard ASP.NET Core and SQL Server, so the skills to maintain it are widely available locally. Deployable on-premise or in any cloud, so no dependency on a single foreign provider. No per-seat licence cost as the entity count grows. The department can take the codebase in-house.

9. Security, privacy and responsible AI

Requirement (e) of the brief, and criterion 6 of the adjudication (10 percent).

Access control

Role-based, configured in Shesha's security model. Four roles:

- **Entity reporter.** Sees and submits only for their own entity.
- **DSAC reviewer.** Sees all entities, reviews and returns submissions, cannot alter submitted figures.
- **DSAC executive.** Read and summary access across the portfolio.
- **Administrator.** User and configuration management, no ability to alter submitted performance data.

The public surface is a separate unauthenticated projection exposing only a deliberately narrow, non-personal subset.

Data protection

POPIA-aligned. Personal information in scope is limited: reporter contact details and aggregated workforce demographics. Demographics are held and displayed in aggregate only, never at individual level.

Encryption in transit and at rest. Full audit trail on every change to a submission, target or document, with actor and timestamp. Data residency in South Africa, which the deployment model permits by default.

Responsible AI

Three commitments, each enforced by the architecture rather than by policy alone.

Nothing machine-derived enters the record unconfirmed. Extraction results live in a separate table with confidence and source location. A named human confirms them into TargetResult. The separation is structural, not procedural.

Every figure retains its provenance. Each extracted value points back to the document and location it came from. A reviewer can always open the source.

Risk scores are decomposable and deterministic. The score is rule-weighted, not a trained black box. The interface always shows the contributing signals. A reviewer can disagree with any one of them, and the reasoning is auditable after the fact.

Deliberate limits

The system does not auto-approve submissions, does not publish anything to the public surface that has not been through DSAC review, and does not generate performance judgements presented as fact. It surfaces signals; people decide.

10. Judging criteria mapping

#	Criterion	Weight	What earns it
1	Relevance to the challenge	20%	All five mandatory requirements built (\$4). Partial implementation is explicitly allowed where the approach is credible, so anything descoped is presented with a concrete path.
2	Innovation and emerging technologies	15%	Document extraction, behavioural risk prediction, WhatsApp submission channel. The accountability-chain reframe itself. Indigenous capability via an SA open-source stack.
3	Technical feasibility and functionality	20%	A working end-to-end path, not slides. Submission actually flows to dashboard to public view in the demo. Integration capability shown through the API layer.
4	UX, accessibility, inclusivity	10%	WhatsApp/SMS channel for under-resourced NPOs, low-bandwidth public view, plain language, mobile-first. A channel, not a font-size claim.
5	Data, intelligence, insight	10%	Rand per outcome. Risk scores displayed with contributing signals. Every figure traceable to source. Transparency and explainability are structural.
6	Security, governance, responsible tech	10%	Role-based access, POPIA alignment, audit trails, human-in-the-loop enforced by the data model, deterministic explainable scoring.

#	Criterion	Weight	What earns it
7	Scalability, sustainability, sovereignty	10%	Open-source SA framework, standard .NET and SQL Server skills, on-premise or any cloud, no per-seat licensing, department can take it in-house.
8	Presentation and demonstration	5%	The 90-second thread in §6.5. One story, three audiences, no slide needed.

Where the points actually are

Only 20 percent is tied to the specific challenge statement. Another 45 percent — innovation, data and insight, security, sovereignty — rewards how we build and how we explain it, and is available to any team that thinks about it. Most will not.

The implication for the build: get the mandatory floor working first, because criteria 1 and 3 together are 40 percent and both require things that visibly function. Then spend remaining time on explainability and the public view, which are cheap to build and carry disproportionate weight.

11. Build plan and cut list

The build window is not yet confirmed. The tiering below holds regardless: build strictly in order, and cut from the bottom.

Tier 1 — must exist or we have no submission

1. Domain model implemented, CRUD APIs generated, seed data for a handful of entities with realistic targets and a few years of history. **Realistic seed data matters more than people expect**; the demo is only convincing if the numbers look like real APP targets.
2. Entity submission flow: upload a document, confirm structured results, submit.
3. DSAC dashboard: entities ranked, targets by state, drill into one entity.
4. Document repository with versioning and approval state.
5. Countdown notifications at 30 days, 15 days, hourly.
6. Role-based access working across the three roles.

Tier 2 — what makes us win rather than place

7. Risk score with visible contributing signals.
8. Document extraction, even if narrow. Handling one document layout well beats handling four badly.

9. Rand-per-outcome comparison view.
10. Public entity page.

Tier 3 — the flourish

11. WhatsApp or SMS submission channel.
12. Report-once export in a second format.
13. Executive narrative summary.
14. Comments anchored to targets, task assignment.

Cut list, in the order things get dropped

- **Microsoft 365 integration** → becomes a roadmap slide with an Entra ID and Graph approach described.
- **Real-time comment sync** → becomes threaded comments with refresh.
- **Report-once second format** → becomes a described capability of the data model.
- **Executive summary generation** → cut entirely; least visible in a 90-second demo.
- **WhatsApp channel** → if it must go, replace with a low-bandwidth mobile web view and explain the channel as the next step. Painful, since it is the strongest inclusivity story we have.

Non-negotiable

Whatever else is cut, the demo thread in §6.5 must run end to end without a stub. A judge forgives missing features; they do not forgive a demo that does not work.

Suggested split

- **Domain and backend:** model, APIs, seed data, notification and scoring jobs.
- **Forms and dashboards:** everything achievable in the form builder, which is most of Tier 1.
- **Custom components:** risk panel, comparison view, public page.
- **Extraction and channels:** extraction service, then WhatsApp webhook.
- **Pitch and demo:** owned by someone from the start, rehearsed at least three times, not assembled in the last hour.

12. Open questions and assumptions

Needs answering before the build starts

How many build hours do we actually have? Determines where the line falls between Tier 2 and Tier 3.

Team size and skill split. How many are comfortable in .NET, how many in React, how many can drive the Shesha form builder productively on day one.

Is there a Shesha starter template provided for the hackathon, or do we set up from the public repo? Setup time is dead time and should be established immediately.

Can we get real or realistic DSAC data? Published annual reports from these entities are public. A few hours spent pulling real targets and real allocations from two or three published APPs would make the demo dramatically more convincing than invented numbers.

Is there a Boxfusion mentor available? They said to come find them. Use that early for the framework's sharp edges rather than late for debugging.

Assumptions made in this document

- Shesha is mandatory for the build, not merely encouraged.
- No live integration to any DSAC or Treasury system is available; everything is seeded or uploaded.
- The extraction target is documents we supply as samples, not arbitrary real-world APPs.
- "Full functionality on mobile devices" in requirement (d) is satisfiable by responsive web, not a native app.
- Judging is a live demo plus questions, so the presentation format assumed in §6.5 holds.

Deliberately out of scope

- Financial system integration or transaction-level expenditure tracking
- Any workflow that certifies or signs off performance information for audit purposes
- Multi-department extension beyond DSAC, though the model supports it and that is worth saying aloud